

A null model of cultural spread may explain the long-standing puzzle of simultaneous inventions

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Abstract

Simultaneous inventions, or *simultaneous discoveries*, describe the phenomenon that when analogous developments occur independently, they are often clustered within a short timeframe. Thus, for example, Leibniz and Newton famously happened to formulate calculus independently at virtually the same time. This has fascinated scholars and the general public for years, and different explanations have been offered, usually invoking the idea of the *Zeitgeist*, "Spirit of the age." It suggests that the concurrent occurrence results from external circumstances that render the inventions likely only at particular time windows. We propose a simpler, more parsimonious explanation that makes no such assumptions. We suggest that even without any change in external conditions, analogous inventions should be expected only when they occur closely in time. This is because one cannot independently invent something that they already know, and thus, once an invention occurs, a second independent invention can occur only within a short time window prior to the complete spread of the first. We develop a model of cultural transmission inspired by genetic soft sweeps – the spread of independent analogous genetic adaptations – that explores this phenomenon. We show that the faster the cultural transmission – whether because of the characteristics of the invention, the population, or the inventor, the shorter the time window for analogous inventions to appear. Finally, we suggest that even when the intrinsic probability of invention is uniform across space, a positive association between the geographic proximity of the simultaneous inventions and their proximity in time is to be expected.

Introduction

For over a century, people have been puzzled by the phenomenon of simultaneous inventions and discoveries (Ogburn & Thomas, 1922; Merton, 1963; Brannigan et al., 1983; Gratzinger, 2011; Shama, 2019; Kokol et al., 2020). This phenomenon, where the origins of the same cultural advancement found in a population can be traced back to multiple independent sources, all emerging around the same time, challenges our understanding of innovation. If discoveries/inventions are a product of “geniuses” of exceptional abilities, how come these geniuses often happen to live so closely in time to one another? Alternatively, if such advancements are inevitable, why would they appear in such narrow time windows?

From the independent emergence of agriculture within several thousands of years in multiple different regions in the Old World, and similarly, in several regions in the New World (Larson et al., 2014) to the independent discoveries of touch and temperature receptors (Caterina et al., 1997; Story et al., 2003) that won the Nobel Prize in Physiology or Medicine in 2021, such multiple inventions can be found in almost every period, geographical region and cultural domain.

While many have agreed that this phenomenon is surprising, it is important to clearly state what the non-intuitive observation is that requires explanation. The observation we tackle here is that when analogous inventions or discoveries (i.e., sharing the same function or meaning but being of independent origins) occur and become established (i.e., are transmitted to a non-negligible fraction of a population), it often happens very closely in time (thus “simultaneous”). The reason this observation is surprising is that given the naïve expectation that analogous inventions are indeed completely independent of one another, they should be evenly distributed throughout time and not correlated to one another. Note, again, that the term “invention” in this context refers not only to the occurrence of the idea or discovery in one’s mind but also to its cultural spread. We henceforth use “simultaneous inventions” to refer to analogous inventions or discoveries that happened within a short time window, and “double inventions” to any pair of analogous inventions.

The prominent theory in explaining this phenomenon is the *Zeitgeist* ("Spirit of the age", e.g., Ogburn and Thomas 1922). According to this theory, simultaneous inventions occur due to certain sociocultural or environmental circumstances that make them possible, or likely only at a specific period. For example, some inventions/discoveries rely on others and, therefore, cannot precede them. The simultaneous discovery of sunspots, for instance, by Thomas Harriot

(1610), Johannes and David Fabricius (1611), Galileo Galilei (1612), and Christoph Scheiner (1612) became possible only after the invention of the telescope at the beginning of the 17th century (Brody, 2002). Once telescopes existed, it was perhaps expected for multiple people to use them to explore the sun and, hence, increase the probability that they would identify the same phenomenon. Other sociocultural factors that allow for inventions may not be as conspicuous. For example, several different attributes of Victorian Britain have been suggested as enabling the independent development of evolutionary theory by Charles Darwin and Alfred Russel Wallace in the mid-19th century. These range from similar preexisting concepts to imperialism and the exploration of new ecosystems (Fichman, 2010). Either way, the *Zeitgeist* theory suggests that simultaneous inventions are, in fact, not entirely independent, as they result from shared conditions across origins.

Here, we propose a null model for the proximity in time of analogous inventions. While **not ruling out** the contribution of a *Zeitgeist* in some cases, we suggest that the clustering in time of analogous inventions is to be expected simply due to the cultural dynamics of the spread of knowledge. This study thus offers a simple explanation that does not rule out alternative explanations, but that should – in the spirit of Occam’s razor (Sober, 2015) and the principle of parsimony as portrayed by Popper (Popper, 1959) – be preferred.

The conceptual model

We suggest that once an invention occurs and begins to spread, there is only a limited time window for analogous inventions to emerge. This is because, by definition, an independent invention can only occur and become established if it is invented by a naïve individual and transmitted to their similarly naïve immediate social circle. As inventions spread, the number of naïve individuals, i.e. potential inventors, decreases, and so does the probability of additional independent invention events. Using a simple spatially implicit model, and drawing inspiration from the mathematical theory on soft selective sweeps (i.e., the phenomenon where multiple analogous genetic adaptations spread simultaneously; Hermisson & Pennings, 2005; Pennings & Hermisson, 2006; Ralph & Coop, 2010; Messer & Petrov, 2013), we show how cultural transmission – in this case, the spread of inventions, or knowledge of their existence – determines the probability of multiple analogous inventions and constrains the time window for their occurrence (for elaboration on the mathematical model, see Methods). Afterward, we

explore how incorporating spatial structure can enhance our understanding of simultaneous inventions.

Results

To formally explore the effect of the transmission of knowledge on the clustering in time of analogous inventions, we use a simple mathematical model (see Methods). We use a set of assumptions that is common in similar models of cultural evolution (e.g., Henrich 2001, Kolodny et al. 2015) and population genetics (e.g., Hermisson and Pennings, 2005; Pennings & Hermisson, 2006; Ralph & Coop, 2010; Messer & Petrov, 2013) to allow for meaningful comparison, integration, and the transfer of insights between models. Importantly, however, the model is robust to deviation from nearly all of its assumptions, always predicting that analogous inventions would be clustered closely in time. The only crucial assumption is (the trivial one) that by definition, an invention cannot be considered an “independent invention” if the candidate inventor or the society around them is already aware of the existence of the invention.

We model a population of N naïve individuals (unexposed to the invention), where each individual has an intrinsic probability of invention μ . Once an invention occurs, it starts spreading in the population as a function of a transmission coefficient r , following the logarithmic growth equation:

$$(1) \quad \frac{df}{dt} = r \cdot f \cdot (1 - f),$$

Where f is the fraction of the population exposed to the invention ("demonstrators"), as a function of time t . After an invention first occurs, the fraction of demonstrators gradually increases, and correspondingly, the fraction of naïve individuals (potential inventors), decreases. As a function of the reduction in the number of potential inventors, the probability of additional inventions declines and reaches zero when the entire population is exposed to the invention.

Using this simple set of assumptions, we can now demonstrate how different factors are expected to affect the number of inventions and their proximity in time, using computer simulations with stochastic invention events.

1. We begin by exploring the effect of the transmission rate (r) on the number of inventions and their proximity in time. For a population of 1000 individuals, and an individual invention rate (μ) of 10^{-6} , we ran a series of simulations with r ranging between 0 and 0.09 for 1000 time units (see Figure 1). As expected by the analytical calculations in the Methods section, we show that higher r values are associated with a lower number of analogous inventions. More importantly, we find that when multiple analogous inventions do occur in simulation runs with high r , they are always clustered in the short time window between invention and fixation (i.e., $f = 1$, when the entire population is exposed to the invention). The higher the rate of transmission, the more closely clustered inventions are in the cases of double inventions.

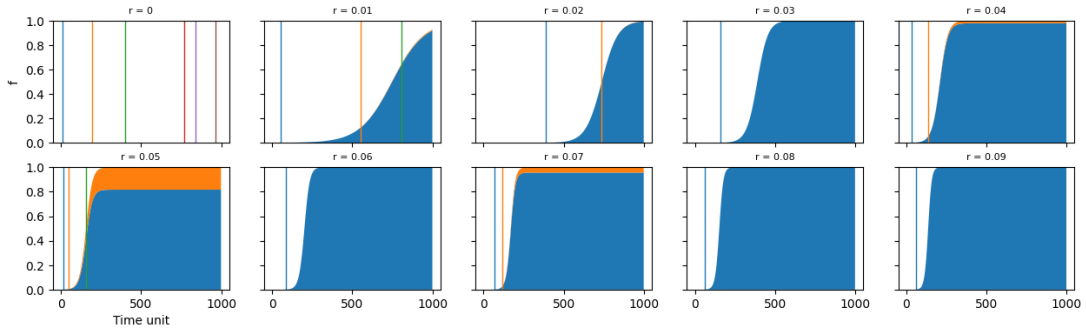


Figure 1 The faster the cultural transmission, the fewer the analogous inventions on average, and the closer they are temporally. Here, each graph represents a simulation run for the same population characteristics ($N = 1000$, $\mu = 10^{-6}$) with different transmission coefficients (r values). In each simulation run, colored vertical lines illustrate analogous inventions, and the proportion of the population exposed to the invention – f is monitored over time, broken down to transmission lineages, corresponding in color to their origin (Blue for the first invention, orange for the second, etc.). We find that the higher the r , in general, the fewer the analogous inventions, but also that when multiple analogous inventions do occur, they are necessarily close in time (compare, for example, the results for $r = 0.01$, $r = 0.07$ and $r = 0.08$, which happened to result in two invention events).

2. Next, we illustrate a potential pitfall that may emerge due to the dynamics of analogous inventions, if such data is used to infer the probability of invention. As explained above, analogous inventions are observed only in the (possibly rare) cases when they happen to occur closely and early enough in time before the original invention takes over the population. In other words, the time intervals between such inventions may be non-characteristic of the per-time-unit probability of invention. To demonstrate this, we explore a scenario with a population of size 1000, a high transmission rate ($r = 0.05$), and a low invention rate ($\mu = 10^{-6}$) where analogous inventions are unlikely to occur (the expected number of inventions is 1.14 per simulation run; see Methods for

calculation). For this scenario, we ran the simulation a million times, 24 of which are presented in Figure 2. A total of 1.384 million inventions occurred overall, meaning that the average number of inventions per simulation was 1.384, very close to the expected value. Of the million simulations, 129,229 ended with more than one invention occurring, and a total of 138,357 analogous inventions were recorded (these numbers differ from one another because in some cases, more than one analogous invention followed the original). The average interval between analogous inventions was 72 time steps. If one were to ignore the spread of knowledge and try to infer the intrinsic probability of invention from these cases only (i.e., ask at which value of μ would we expect an additional invention to occur within 72 time units for a population of size 1000), they would estimate it as 1.39×10^{-5} , almost 14 times higher than the actual value of μ in the model. In addition, the cases in which analogous inventions are observed, under this set of conditions, are usually characterized by a long period of no inventions (1000 time units, on average), followed by two or more closely clustered inventions. In other words, looking only at the non-representative cases in which analogous inventions do happen to occur, can lead to the impression of a high short-term probability of invention (a *Zeitgeist*.)

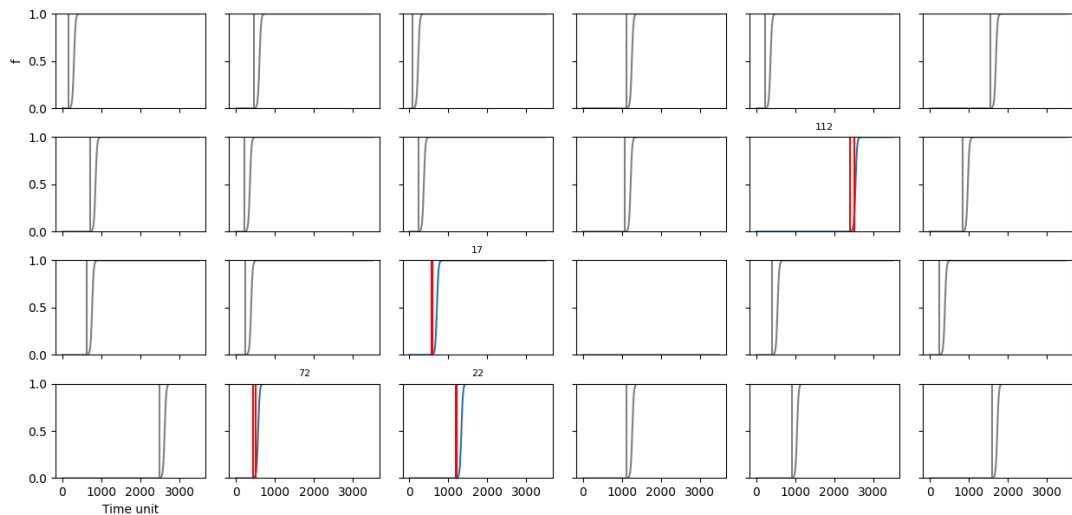


Figure 2 When the spread of knowledge is rapid, analogous inventions are less likely, but when they occur, they are clustered in time. Here, 24 simulation runs (out of a total of 1 million) are shown for the same parameter combination. Multiple analogous inventions are unlikely under these parameters ($N = 1000$, $\mu = 10^{-6}$, $r = 0.05$): the expected number of inventions is 1.14. Thus, in most cases (panels depicted in greyscale), the original invention fixates before any analogous invention emerges. When analogous inventions did occur (where f is shown

in blue, and inventions are marked by red vertical lines), the average interval between them was 72 time units, as opposed to the average of 1000 time units it took the invention to occur in a naïve population.

3. Finally, while in the previous examples, we assumed no population structure for simplicity, we now explore the effect of spatial structure on the occurrence of analogous inventions (see Figure 3). For the ease of visual representation, we illustrate the spread of inventions in a homogeneous space, but we emphasize that in real life, the spread of cultural knowledge is a product of social network structure, which does not necessarily correlate with geography. Thus, “distance between individuals”, as discussed below, is defined as the distance within the social transmission network. Consider that an invention happens at some point in space and starts spreading in a homogenous space. The greater the distance from the invention location, the longer it would take for the invention to spread. This means that distant individuals have a longer time to come up with an analogous invention before the original invention reaches them. Therefore, when analogous inventions are found very closely in space, it is necessarily because they happened closely in time, while when analogous inventions are very distant from one another, there could be a much larger difference in the timing of their occurrence.

In Figure 3, we illustrate the spread of an invention in space over time, borrowing the graphic representation of papers on soft sweeps (e.g., Ralph and Coop 2010). Looking at Figure 3B, imagine we would randomly place inventions in time and space. Because analogous inventions are possible only in the white areas (as the red area represents the spread of the original invention), there will be **a positive correlation between the proximity in time and the proximity in space of analogous inventions** (for a detailed demonstration, see section S3 in the Supplementary Materials).

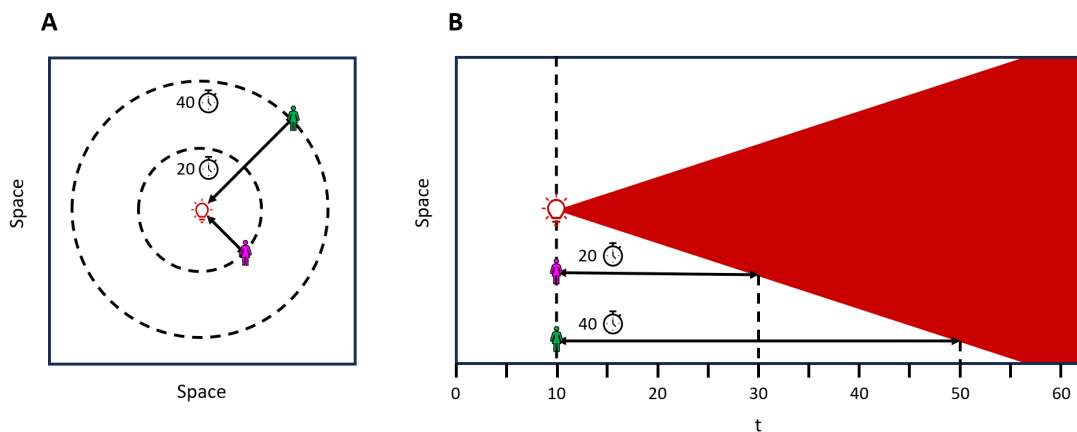


Figure 3 Transmission distance sets the boundary for the distance in time between inventions. Panel A illustrates the spread of an invention in a two-dimensional space, and Panel B illustrates the same process in a single-dimensional space represented by the y-axis, where the x-axis represents the progress in time. The lightbulbs represent the original invention event, and from them, the invention spreads at a constant rate, to all directions in Panel A, and upwards and downwards on the vertical line in Panel B. In panel B, the shaded area represents the portion of the vertical line to which the invention has already spread at a given time. Under these conditions, the purple individual has up to 20 time units to come up with an analogous invention before the original invention reaches them, while the green individual, who is more distant, has up to 40 time units. This shows that whenever analogous inventions happen in proximity (geographic, social, etc.), they will necessarily (in this simplified model) be close in time.

Discussion

In this work, we address the age-old puzzle of simultaneous inventions, suggesting a parsimonious explanation compared to the common *Zeitgeist* interpretation. Notably, although Occam's razor suggests that the parsimonious explanation generally be preferred, in this case the two explanations are not mutually exclusive, and there are likely to have been many cases in which both were at play. Our model sheds light on a trivial yet underappreciated aspect of analogous inventions: the fact that a second invention can only be observed if it happens to occur at a time and place where the original invention is yet unknown. Thus, considering the spread of knowledge, analogous inventions are observed only when simultaneous. An analogous invention has to occur, by definition, after the original invention appeared and before it takes over the entire population. The distance in time between these two events is determined by the pace of cultural transmission.

Within the time window between the original invention and the invention's fixation, the probability of additional inventions gradually declines as more and more individuals are exposed to the invention. For instance, while there are probably still some people around the world who are naïve to wheels, their number is extremely low, making the reinvention of the wheel today extremely unlikely.

Importantly, simultaneous inventions are often characterized not only by their clustering in time but also by the extended periods of time before they begin to occur, which further contributes to the impression of a *Zeitgeist*. As explained above, we suggest that a *Zeitgeist* may indeed facilitate this observation in some cases. However, our results show (Figure 2) that even without a *Zeitgeist*, but rather under **constant** low probabilities of invention, whenever

analogous inventions are seen, they tend to follow this same pattern. Namely, long periods during which the invention is possible but does not occur, followed by a series of two or more clustered invention events.

Over the years, studies have analyzed real-life cases of analogous inventions/discoveries, trying to infer the probabilities of certain inventions or the time at which they became possible (e.g., Brannigan and Wanner, 1983; Key et al., 2021). Because of the spread of knowledge, analogous inventions will be observed only if they happened to occur nearly simultaneously. This may create a misleading impression of a temporary increase in the probability of invention; in other words, it may lead to unwarranted assertions regarding the existence of a *Zeitgeist*. Complementarily, ignoring the role of the spread of knowledge in prohibiting reinventions may also lead to an underestimation of the intrinsic probability of invention μ , since even inventions that are extremely likely may occur no more than a few times due to the spread of knowledge.

Finally, we use a simple thought experiment to demonstrate how population structure is expected to affect the occurrence of analogous inventions, arguing that the null hypothesis in analyzing the relationship between distance and invention time should be a positive correlation. That is, inventions that happened to occur closely in space necessarily happened nearly simultaneously, while inventions that occurred in distant locations could occur either closely or distantly in time (and, on average, later). Consider the emergence of agriculture as an example. It is known that agriculture was developed independently in multiple centers worldwide (Larson et al., 2014) and that whenever it emerged, it spread rapidly to nearby locations. Therefore, when we find multiple independent centers near each other, as, for example, is the case in the Americas, the expectation should be that the invention of agriculture occurred at these centers almost simultaneously, regardless of whether the conditions in the Americas at that time were unique in any way.

Previous research (Makari, 2015; Grayling, 2016) emphasized that analogous inventions seem to become more prevalent over time. The common interpretation of this observation is that it is a product of increased interconnectivity in modern periods, which arguably makes the *Zeitgeist* (i.e., the probability of invention) more coordinated across populations. At first glance, this observation may seem inconsistent with our model's predictions, as the efficient transmission of knowledge is expected to shorten the time window for analogous inventions, making them less likely. We suggest, however, that there are other factors that may contribute to the observed increase in simultaneous inventions over time. First, modern times are not only

characterized by increased connectivity but also by increased rates of inventions, which are expected, according to our model, to increase the probability of analogous inventions. Second, the observed surge in simultaneous inventions may also be a product of our better ability to detect them in more recent periods. Simultaneous inventions can go unnoticed by researchers for different reasons: Because they are difficult to distinguish from one another (particularly if close in time and space), due to lack of documentation, or because, over time, one of them dominated the population stochastically or due to its superior functionality. The further we look into the past, the lower the resolution of the historical/archeological record. This raises the likelihood of earlier analogous inventions going unnoticed, and our ability to differentiate between convergent vs. single-origin inventions is highly compromised.

In our model, the speed at which inventions spread in the population is determined by a single parameter, r . In reality, the rate of cultural transmission is a product of many factors, such as the rate of connectivity in the population, the fitness of the cultural trait (which in itself is a product of functionality, attractiveness, etc.), the rate of conservatism/openness to new ideas and even the identity and prestige of the inventor. While each of these factors can be further investigated, we predict that low connectivity, low functionality, low attractiveness, and high conservatism are likely to decrease r and, therefore, increase the probability of analogous inventions. It should be noted that, however, these factors are expected to affect the population in aspects other than its r . It is possible, for example, that highly conservative populations, expected to show low r , would have a lower μ , which might actually reduce the probability of analogous inventions. Similarly, as suggested by the *Zeitgeist* theory, inventions are commonly enabled by certain sociocultural circumstances, which may be less likely to be the same in loosely connected populations. In such a case, inventions would be possible only in a fraction of the population, making analogous inventions less likely.

This model was designed to show a few simple examples of the effect of cultural transmission on the occurrence of simultaneous inventions. It should be noted, however, that population genetics offers an extensive theoretical framework that can easily be adapted to investigate more complex scenarios. In fact, it even discusses several mechanisms that are analogous to a *Zeitgeist* – changes in the genetic architecture that make adaptations possible, changes in selection pressures affecting the survival and spread of adaptations, and changes in the probability of mutation, shaping directly the probability of adaptation.

As mentioned above, a genetic phenomenon that is particularly similar to analogous inventions is soft selective sweeps. Soft sweeps are cases where analogous beneficial mutations of different sources are established together in a population and fixate. These mutations can appear either before or after their favorable selection begins, with the latter (sometimes referred to as soft sweeps type II) sharing more similarities with our common conception of simultaneous inventions. Several studies have developed more complex models to explore a wide range of scenarios which may be valuable for the study of analogous inventions. For example, they explore the effect of different population structures and demographic dynamics, such as migration between partially connected populations (which is of particular interest in the study of cultural evolution; e.g., see Derex & Boyd 2016; Derex et al., 2018; Ben-Oren et al., 2023a; Ben-Oren et al. 2023b, Ben-Oren et al. in prep) on the probability of soft sweeps (e.g., Ralph and Coop 2010; Messer & Petrov, 2013).

Another important feature explored by these studies is the effect of stochasticity on the establishment of adaptive mutations (equivalent to inventions in our study). While we assumed, for simplicity, that inventions spread deterministically and unidirectionally, in reality, inventions may commonly fail to establish, regardless of their value. Similarly, even when analogous inventions do occur, some “lineages” may be lost stochastically over time, not having descendants in the present.

One scenario, for example, that we believe deserves further investigation is a non-symmetrical spread of knowledge between different parts of the population. Imagine, for instance, a population that has a "cultural core" from which cultural traits are frequently transmitted to other parts but not vice versa. Inventions that happen first outside of this core are unlikely to penetrate it, increasing its chances of independent invention. Inventions that happen in the core, on the other hand, are expected to rapidly spread to the rest of the population, preventing additional inventions. Such conditions can create the impression of higher innovativeness in the core, even if all individuals have the same probability of invention (μ).

As mentioned above, the type of soft sweeps that we find particularly relevant for the modeling of simultaneous inventions is soft sweeps type II (where, due to a certain selection pressure, multiple novel analogous adaptations appear and spread in parallel). However, another important dynamic explored by models of selective sweeps is a temporal change in selection pressures, for example, when selection only begins at a certain time point. Under neutral conditions, i.e., without directional selection/spread, the same adaptation/invention can occur

numerous times without spreading to high frequency (e.g., as may have been the case with symbolism, Hovers and Belfer-Cohen 2022; Ben-Oren et al., 2025), either surviving persistently at low frequencies or being lost stochastically. Then, if the selection regime changes, preexisting and/or novel inventions may suddenly start spreading, closing the window for additional inventions.

In conclusion, while not contradicting the potential role of a *Zeitgeist*, our results suggest that the fact that analogous inventions are commonly clustered in time and space does not necessarily mean there was anything unique about the circumstances of their appearance. We believe that accounting for the effect of cultural transmission in constraining analogous inventions will profoundly impact our interpretation of the phenomenon of simultaneous inventions.

Methods

Mathematical Framework

We develop a simple mathematical model for the emergence and spread of inventions, inspired by similar models of cultural evolution (e.g., Kolodny et al., 2015) and population genetics (e.g., Hermisson & Pennings, 2005; Pennings & Hermisson, 2006; Ralph & Coop, 2010; Messer & Petrov, 2013) exploring the spread of adaptive alleles/cultural innovations.

We assume that for a population of N naïve individuals (unexposed to the invention), where each individual has an intrinsic probability of invention μ , the probability of a new invention per time unit is $N \times \mu$. For simplicity, our model is deterministic, meaning that whenever an invention occurs, it always spreads in the same manner. Once an invention occurs, it starts spreading in the population. In biological models of “well-mixed” populations, the spread of genetic adaptations is primarily a product of their fitness coefficient (i.e., their advantage to the survival and reproduction of the organism). In our case, the spread of a cultural invention is determined by many different features including the functionality and the attractiveness of the invention, as well as the population’s degree of connectivity and openness to change, and could even be affected by different characteristics of the inventors themselves. All these characteristics are encompassed in a transmission coefficient r , representing the transmission rate in the population. For simplicity we chose to implement cultural transmission using the simplest commonly used model; it is assumed to follow the logarithmic growth equation:

361 (1) $\frac{df}{dt} = r \cdot f \cdot (1 - f),$

362 Where f is the fraction of the population exposed to the invention ("demonstrators"), as a
 363 function of time t . This equation is commonly used in modeling both biological and cultural
 364 transmission dynamics (Henrich, 2001; Otto & Day, 2007), and the idea behind it is that
 365 transmission is impacted by the fraction of "infected" individuals / cultural "demonstrators" in
 366 two ways. On one hand, the more demonstrators, the more sources there are for transmission.
 367 On the other hand, the larger the fraction of demonstrators, the less naïve individuals are left
 368 to transmit the invention to.

369 While this model will make some simplifications and assumptions on the way cultural
 370 inventions occur and spread (e.g., that the probability of invention is equal across all individuals
 371 or that the spread of invention follows the logarithmic growth equation), it is important to note
 372 that these are only used to make the model simple and tractable and are not requisites for the
 373 proposed dynamics that lead to simultaneous inventions.

374 Additionally, our model is deterministic, meaning that whenever an invention occurs, it always
 375 spreads in the same manner. However, regardless of the fact that, in reality, many inventions
 376 (similarly to genetic adaptations) may be lost due to stochasticity, a positive relationship
 377 between population size and the emergence of inventions is expected.

378 Assuming that the invention is generated by one individual in a population of size N at time t
 379 $= 0$, the fraction of demonstrators, f , can be represented by:

380 (2) $f(t) = \frac{\frac{1}{n}e^{rt}}{1 + \frac{1}{n}e^{rt}}$

381 For full mathematical derivation of this equation, as well as the following ones, see
 382 Supplementary Materials. A graphical representation of Eq. 2 shows that the spread of cultural
 383 innovation is expected to follow a sigmoid curve:

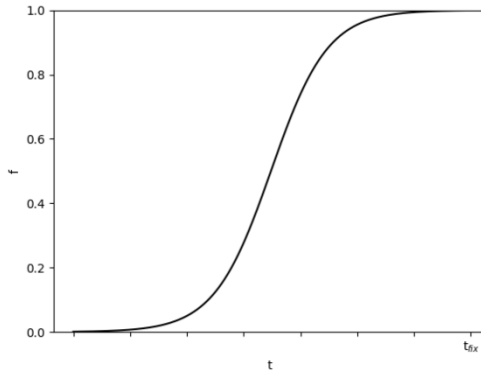


Figure 4 The spread of inventions is expected to follow a sigmoid curve. The x-axis shows the progress in time, and the y-axis shows the fraction of individuals exposed to the invention.

Figure 4 shows the spread rate is initially slow since only a few demonstrators can transmit the invention. Afterward, this rate accelerates due to the growing number of demonstrators and because the number of naïve individuals to whom the invention can be transmitted is still large. Finally, after most of the population is exposed to the invention (which happens after $t = t_{fix}/2$), it slows down.

Given a population of size N , and a transmission coefficient r , we can estimate the time window in which, following the initial invention, analogous inventions may occur, that is, the time between invention and fixation (i.e., the time by which all individuals had been exposed to the invention):

$$(3) \quad t_{fix} = \frac{2 \ln(N-1)}{r}$$

The equation shows that, as intuitively expected, the smaller the population and the faster the transmission, the quicker the invention will spread to the entire population.

Based on the approximation of Eq. 2, we can now calculate the probability of reinvention p in the population, per time unit:

$$(4) \quad p(t) = f(t)N\mu = \frac{\frac{1}{n}e^{rt}}{1 + \frac{1}{n}e^{rt}}N\mu$$

As invention spreads in the population, the probability of additional reinventions starts to decrease. This is because once an individual is exposed to the invention, it cannot reinvent it. Before an invention occurs, for a population size $N = 1000$ individuals and a probability of invention $\mu = 10^{-6}$ per individual, the probability of invention is roughly $N\mu = 10^{-3}$ per time unit (Figure 5). When the invention spreads to half of the population, there are only 500

individuals left who can reinvent it independently, reducing the probability of invention to 0.5×10^{-3} per time unit. When 100 naïve individuals are left, this probability reduces to 10^{-4} , and when 10 are left, it reduces to 10^{-5} . Finally, when the entire population is exposed to the invention, the window for analogous inventions is closed, and no additional reinventions may occur.

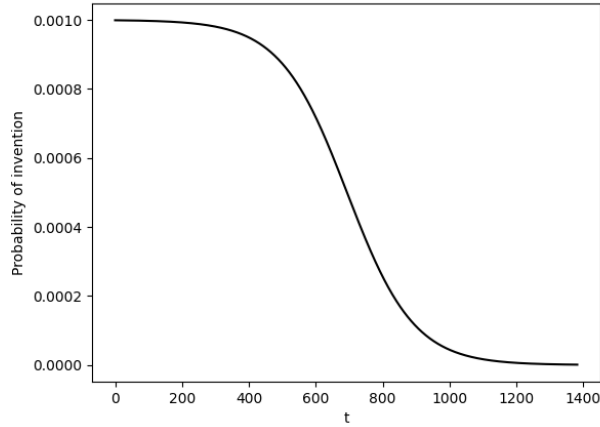


Figure 5 As an invention spreads in the population, the probability of additional analogous inventions decreases. The x-axis describes the time t , and the y-axis describes the probability of an analogous invention to occur per time unit for a population of $N = 1000$ individuals, an individual probability of invention of 10^{-6} per time unit, and a transmission rate r of 0.001 (see Eq. 4). The figure shows that right after the first invention occurs, at time $t = 0$, the probability of an analogous invention is $\sim 10^{-3}$ per time unit, and as the invention spreads, it gradually decreases to zero.

While many well-known examples of simultaneous inventions consist of only two invention instances, this number may be, and occasionally is, larger. Using the equations for the time of fixation and the equation for the probability of invention per time unit, we can estimate the number of independent inventions expected before fixation λ as:

$$(5) \quad \lambda = 1 + \frac{\ln(N-1)N\mu}{r}$$

The rationale here is that once an invention appears and starts spreading, the number of additional inventions would be a function of the number of time units before fixation and the average probability of invention per time unit (which is $(N/2) \times \mu$, since the spread function is symmetrical and averages at 0.5, meaning that across this process, half of the population is naïve on average; See Supplementary Materials section S2 for details).

Based on these equations described above, we can now demonstrate how different factors are expected to affect the number of inventions and their proximity in time, using computer simulations with stochastic invention events.

1. We begin by exploring the effect of the transmission coefficient r on the invention patterns. Using the same default parameters (population size $N = 1000$; individual probability of invention $\mu = 10^{-6}$) we run individual simulations with different values of r (ranging from 0 to 0.09) to see how it affects the number of inventions observed and their spread in time.
2. Then, we explore a specific set of parameters under which multiple inventions are unlikely (which happens when cultural transmission is relatively rapid, and the rate of inventions is relatively low). Under these conditions, most simulations are expected to end with a single invention event. However, even in such scenarios, some simulations will result in multiple analogous inventions. We focus on these cases only, showing how doing this could bias our interpretations of the probability of invention and whether this probability changes over time, leading to unwarranted inference of the existence of a Zeitgeist.
3. Finally, using a simple thought experiment, we discuss how population structure can affect the invention/transmission dynamics.

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Supplementary Materials

S1: Estimating the time frame between invention and fixation

The expansion of an invention is expected to follow the logistic growth equation:

$$\frac{df}{dt} = r \cdot f \cdot (1 - f)$$

Where f is the fraction of the population exposed to the invention and r is the transmission rate.

We defined $f_{(t=0)} = I/N$, where N is the size of a population of naïve individuals, meaning that at time zero the invention is known only by the inventor. Additionally, we define the fixation time t_{fix} as $f(t_{fix}) = (N-I)/N$, where t_{fix} is the time required for the invention to spread in the whole population, such that, for $t > t_{fix}$, the proportion of naïve individuals that have not encounter the invention is smaller than 1.

Considering the initial condition $f_{(0)} = I/N$, we can calculate the fraction of demonstrators at any point in time, as:

$$f_{(t)} = \frac{\frac{1}{N}e^{rt}}{1 + \frac{1}{N}(e^{rt} - 1)}, \text{ with } t \text{ 'element of' } (0, t_{fix}).$$

We consider than that function $f_{(t)}$ has an axis of symmetry at $t_{fix}/2$, when half of the population is exposed to the invention. We can then calculate the fixation time t_{fix} as:

$$f_{\frac{t_{fix}}{2}} = \frac{\frac{1}{N}e^{r\frac{t_{fix}}{2}}}{1 + \frac{1}{N}(e^{r\frac{t_{fix}}{2}} - 1)} = \frac{1}{2}$$

$$\frac{1}{N}e^{r\frac{t_{fix}}{2}} = \frac{1}{2} \left(1 + \frac{1}{N}(e^{r\frac{t_{fix}}{2}} - 1) \right)$$

$$\frac{1}{N}e^{r\frac{t_{fix}}{2}} = \frac{1}{2} \left(1 + \frac{1}{N}e^{r\frac{t_{fix}}{2}} - \frac{1}{N} \right)$$

$$\frac{1}{2} \times \frac{1}{N}e^{r\frac{t_{fix}}{2}} + \frac{1}{2} \cdot \frac{1}{N} = \frac{1}{2}$$

$$\frac{1}{N}(e^{r\frac{t_{fix}}{2}} + 1) = 1$$

$$e^{r\frac{t_{fix}}{2}} + 1 = N$$

$$e^{r \frac{t_{fix}}{2}} = N - 1$$

$$\frac{t_{fix}}{2} = \frac{1}{r} \cdot \ln (N - 1) = N - 1$$

$$t_{fix} = \frac{2 \ln (N - 1)}{r}$$

S2: Estimating the expected number of inventions during the transmission phase $t = (0, t_{fix})$

For individual invention rate μ (with unit inventions, per individual, per unit time) and a population of naïve individuals of size N , the probability of invention per time unit is roughly: $N \times \mu$. After the first invention event, the invention starts spreading in the population, reducing the number of individuals that can reinvent it, until the invention is known by the whole population. As calculated in S1, for a population of N individuals and a transmission rate r , the expected time until fixation is $t_{fix} = \frac{2 \ln (N-1)}{r}$. Because during the spread of the invention the number of naïve individuals is gradually reduced, the probability of invention in the population per time unit constantly decreases. As we can see in Fig. A, the average fraction of the population exposed to the invention between $t = 0$ and $t = t_{fix}$ is $\frac{1}{2}$. Thus, the average number of naïve individuals during this time interval is $N/2$.

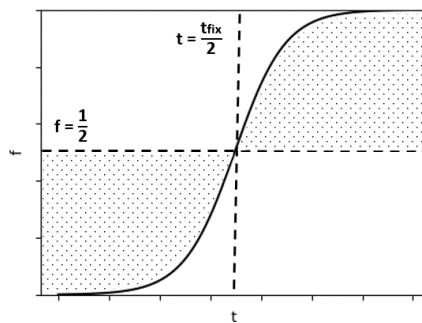


Figure A A demonstration that the average fraction of naïve individuals f throughout the transmission phase is $\frac{1}{2}$.

Thus, the average probability of invention per unit time during the transmission phase can be estimated as $\frac{N}{2} \cdot \mu$, and the expected number of additional analogous inventions a (after the first one occurs), during t element of $(0, t_{fix})$ is:

$$a = \frac{N}{2} \cdot \mu \cdot t_{fix} = \frac{\ln(1-N)N\mu}{r}$$

And adding the first invention event, the total number of expected inventions λ during t element of $(0, t_{fix})$ is:

$$\lambda(N, \mu, r) = 1 + \frac{\ln(N-1)N\mu}{r}$$

A graphical representation of the equation above, in which we explore the effects of N , μ , and r on the expected number of inventions λ occurring during the transmission phase, is provided in Fig. B:

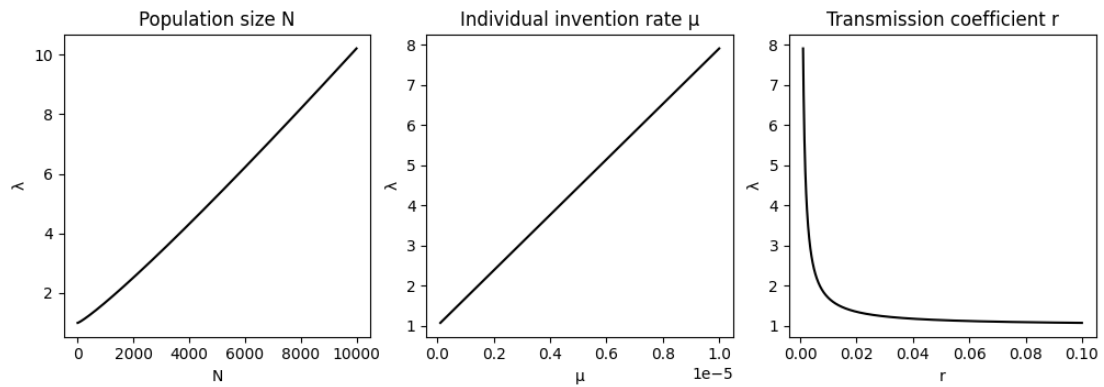


Figure B The expected number of inventions λ as a function of population size (N), individual invention rate (μ) and the transmission coefficient (r) plotted here with default values of $N = 1000$, $\mu = 10^{-6}$, and $r = 0.01$.

We find that, intuitively, higher population size N and higher rate of invention μ lead to an increase in the expected number of analogous inventions, while rapid cultural transmission is associated with low expected numbers of inventions during the time interval $(0, t_{fix})$.

S3: Investigating the relationship the timing of analogous inventions and the time they emerged at

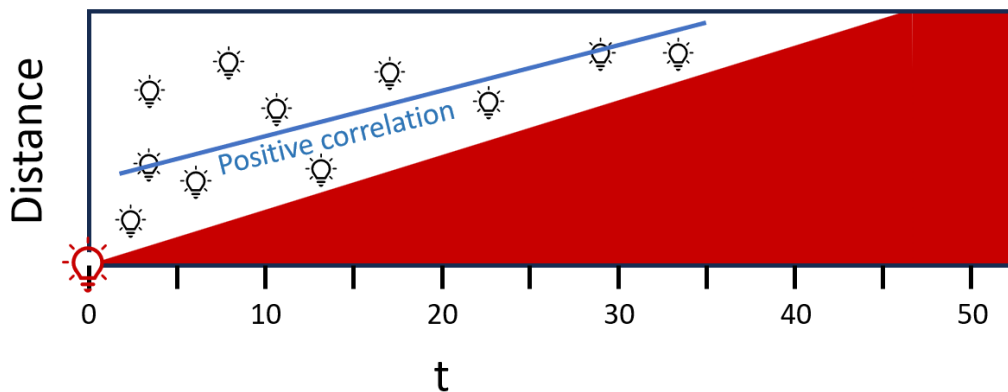


Figure C A demonstration of the positive correlation expected between the distance in time and in space

581 between analogous inventions. Because inventions spread in space, any case where analogous inventions are
582 found closely in space, it means they occurred nearly simultaneously. When analogous inventions are found
583 distantly in space, there is a longer time frame in which they could happen.

584 In this example, the y-axis represents the distance from the point of the first invention,
585 while the x-axis represents the progress in time. The red area signifies the time and places
586 where the original invention already spread, making analogous inventions impossible. If
587 we place analogous invention events randomly in the white space, we will get a positive
588 relationship between invention time and invention place.